



SCHOOL OF POLITICS,
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**Powering the Transition:
Trade Shifts, Critical Minerals, and Policy Dynamics
in the Global EV Market**

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Abstract / Executive Summary

How does the international trade for the market of Electric Vehicles between the EU, the US, and China vary, considering the change in trade flows of crude oil compared to CMs, the infrastructure for EVs, and the impact of government intervention on the EV market through policies and regulations?

This thesis explores the transformative impact of the international trade for the market of Electric Vehicles (EVs) between the EU, the US, and China, given geopolitical frontiers and the allocation of scarce raw materials. With the rise of climate change concerns, the automotive industry is seeing itself in need to gradually change its supply chains from Internal Combustion Engine Vehicles (ICEVs) to EVs, while navigating the challenges posed by regulations and trade tensions, promoting a new era of economic interaction.

The success factors for the EV market to grow are (1) government interventions to ease production and adoption of EVs, which leads to the first hypothesis: *Government policies, including subsidies and tax incentives, significantly influence EV market dynamics by affecting production costs, consumer adoption, and cross-border investment flows*, (2) because the EV market is expected to continue growing, there is a need to develop the appropriate infrastructure to sustain the adoption. Hence the second hypothesis postulates: *The availability of robust EV charging infrastructure is a key factor driving regional EV adoption and trade performance*, and lastly, (3) given the transition towards EVs from ICEVs, the trade flows of critical minerals and crude oil would also be affected. Thus the third hypothesis states: *Shifts in trade flows from crude oil to critical minerals are reshaping global supply chains and trade balances, impacting the international competitiveness of EV industries in the EU, US, and China*. Therefore, this research aims at analysing these market dynamics in function of the main EV industry players, providing recommendations of future steps for all three regions studied.

This study examines through quantitative analysis the key findings, and debates surrounding the influence of the emerging use of critical minerals (CMs), focusing on the roles played by the most prominent global regions in terms of EV production and trade, being the EU, the US, and China. Simultaneously, qualitative analysis delves into the data for the changing dynamics of market structures and the roles of various agents, including EV consumers, car manufacturers, charging station infrastructures, critical minerals implications, and government institutions.

This research contributes to academic literature and provides a foundation for understanding the multifaceted impact of the prices and international trade trends of crude oil and CMs on the growth of the EV market, highlighting the effects of geopolitical matters throughout economies. The study underscores the need for ongoing research to navigate the evolving landscape of EVs influenced by advanced technologies to tackle the scarcity of critical minerals, addressing the challenges posed by the development of battery range, current state of electricity grids, and assessing effectively carbon-neutrality goals.

Key Words: Electric Vehicles, Critical Minerals, International Trade, Policy Analysis, Machine Learning and Time-Series Analysis.

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1. Introduction

The global market for electric vehicles (EVs) is directed towards a steady growth, stimulated by continuous technological advancements, an increasing demand from consumers, and government support, driven by the issues of decarbonisation concerning climate change. Nevertheless, there are still some challenges within this market that need to be addressed for the production of EVs to be fully sustainable and effective, such as improved supply chain mechanisms, inter-regional collaboration for EV infrastructure, and continuous R&D efforts for innovations to remain competitive.

The market for EVs is characterised by aggressive competition streaming from price and technological innovation, especially with regards to the raw materials used for battery production, categorised as critical minerals (CMs), and that impact the final price for the vehicles. The trade balances of the CMs have risen the interest of companies and individuals in the market for EVs, making the shifts in international trade of crude oil for internal combustion engine vehicles (ICEVs) and CMs for EVs a key success factor for the EV industry, together with the state of the infrastructure (such as charging stations), and the impact of policies and regulations which may help or hinder the supply- and demand-sides of this sector. Therefore, this thesis will analyse these market dynamics in function of the main EV industry players, being China, the EU, and the US, answering the research question: *How does the international trade for the market of Electric Vehicles between the EU, the US, and China vary, considering the change in trade flows of crude oil compared to CMs, the infrastructure for EVs, and the impact of government intervention on the EV market through policies and regulations?*

Governments and corporations are aiming to achieve net-zero emissions by 2050 (as for the Paris Agreement), significantly influencing the EV landscape to ensure a sustainable future for the planet and long-term growth, though this requires a significant reshaping of

macroeconomic trade-offs, from short-term economic disruptions versus long-term productivity gains (Hallegatte et al., 2023). Many studies conclude that decarbonisation is the main weapon in order to restabilise the environmental equilibrium across ecosystems, and the optimal way to achieve this is through electrification. This transition from fossil fuels demands massive capital reallocation, approximately \$275 trillion according to a study by McKinsey&Co., to support green energy, EV infrastructure, and sustainable industries (Krishnan et al., 2022). The decarbonisation push reshapes global trade patterns, as countries compete for resource security and supply chain control, and rises geopolitical implications, further highlighting the pivotal role of EVs and its associated technologies (Hallegatte et al., 2023; Mehrotra, 2019).

To clarify the procedure of this investigation: the next section (Section 2) is dedicated to reviewing the literature, delineating existing theories and outlining the hypotheses. Section 3 provides an analytical framework, describing the processes used for the analysis correlating the growth of the EV market to its three major success factors, and presents the limitations of the study. Section 4 includes hypothesis testing, policy analysis, and model analysis, while Section 5 presents a discussion, combining the literature review (2) and the analysis and results (4), and includes potential short-, medium-, and long-term scenarios for the transition to EVs. Finally, Section 6 summarises the conclusion and findings, proposes recommendations for the next steps that the EU, US, and China should take, and suggests future areas of research.

2. Literature Review

2.1. Definitions

For a smoother understanding of the topics, definitions of the key terms used are here provided. EVs are defined by the US Department of Energy as “a vehicle that can be powered by an electric motor that draws electricity from a battery and is capable of being charged from an external source. An EV includes both a vehicle that can only be powered by an electric motor that draws electricity from a battery and a vehicle that can be powered by an electric motor that draws electricity from a battery and by an internal combustion engine (plug-in hybrid electric vehicle)” (US Department Of Energy, 2021). In contrast, an ICEV is a “conventional” vehicle that does not have an electric engine. Nowadays, lithium-ion batteries are the dominant type of rechargeable batteries used in EVs. Battery chemistries depend on five critical minerals designated by the US Geological Survey according to the methodology codified in the Energy Act of 2020 (Congress Research Service, 2022), and their supply is potentially at risk for disruption; lithium (Li), cobalt (Co), manganese (Mn), nickel (Ni), and graphite (C), are the main ones and are will be analysed in this paper.

2.2. Importance of the EV Market

The global EV market is central to economic, environmental, and geopolitical strategies. In fact, in the EU, the US, and China, governments have made commitments to decarbonisation strategies, aiming at reducing greenhouse gas emissions and dependence on fossil fuels, and the EV market is a key driver for this global transition. The EV industry offers businesses both opportunities and challenges: to preserve a competitive advantage, top manufacturers like Tesla, Volkswagen, and General Motors are investing on new EV models, gigafactories, and battery R&D (Hallegatte et al., 2023), and in response, also major Chinese companies like BYD, NIO, and CATL outpaced their Western counterparts, taking the lead in innovation and production (Erbach et al., 2022). Nevertheless, securing CMs supplies remains a major obstacle,

which so far has resulted in an increase in strategic partnerships and government-backed mining initiatives (Network for Greening the Financial System, 2024).

In order to smoothen the transition towards low-carbon economies, all of the three regions adopted measures to benefit the automotive industry through international policy interventions and relative market control. Europe is investing heavily in EV infrastructure, with policies focused on infrastructure expansion and emissions regulations, such as the Green Deal and the Carbon Border Adjustment Mechanism (CBAM). More specifically, the European Green Deal aims to make Europe the first climate-neutral continent by 2050, driving investments into battery production, EV charging infrastructure, and supply chain resilience (Network for Greening the Financial System, 2024), while the CBAM influences international trade by indirectly shaping EV production costs and competitiveness (Mehrotra, 2019). Regarding price and competitiveness, the EU industry in general has higher costs (e.g. energy and labour) than China, so it will probably not compete on lowest price. Despite trailing China, the EU has experienced a sharp growth in EV adoption in the last few years, driven by regulatory frameworks to reduce CO₂ emissions, subsidies for green technologies, and rising investments in battery production (Network for Greening the Financial System, 2024). As an example for smaller car manufacturers in the region, larger automakers such as Volkswagen, Renault, and Stellantis are rapidly expanding their EV production capacity (Erbach et al., 2022).

As opposed to the EU and China, the US lags relatively more in EV infrastructure, translating to barriers to widespread its adoption due to range anxiety and insufficient charging networks. To bridge this gap, the Inflation Reduction Act (IRA) was passed, acting as a major catalyst for both the supply- and demand-sides of the economy, to provide significant monetary incentives for EV production and domestic battery manufacturing, and to increase consumer adoption (Network for Greening the Financial System, 2024). A third benefit of the IRA, is to reduce dependency on China for critical minerals and to establish a robust EV supply chain

within North America, as the reliance on imported CMs and the competition from Chinese manufacturers remain challenges for companies like Tesla, Ford, and GM that are leading the US transition (Erbach et al., 2022). The US has abundant capital, tech innovation, and recently, strong policy support driving its push into critical minerals, yet firms in the US are not the leaders in terms of price competition, as labour and environmental compliance costs are higher, and the industry is relatively new.

Meanwhile, China took the lead in dominating the global market for EV production. As the largest EV producer, China accounts for approximately 60% of global EV sales in 2023 (IEA, 2024a), and dominates the supply chain for batteries and CMs for their production, like lithium, nickel, and cobalt. The Chinese government subsidises domestic car manufacturers, while investing in mining and processing of CMs to secure its competitive edge on the other regions (Mehrotra, 2019). This was possible due to factors ranging from being able to integrate EV production with its economic strategies, leveraging state-led industrial policies, granting aggressive subsidies, and asserting dominance in battery material processing, also because of the location of CMs within its mainland (Hallegatte et al., 2023). China is, as of mid-2020s, the undisputed leader in critical mineral processing and battery manufacturing. This vast refining capacity, estimated 59% of the world's lithium, 73% of cobalt, 68% of nickel, and over 70% of graphite into battery-grade forms (Laje, 2023), combined with large-scale cell production, where over 3/4 of global battery cells are made in China in 2024, gives the country immense leverage and cost advantages. Chinese battery makers benefit from economies of scale and an integrated domestic supply chain, allowing them to offer lower prices, allowing electric cars to get cheaper the fastest, making them highly competitive (Lombardo et al., 2025).

The regions' government policies, together with the proposal in January 2025 to withdraw the IRA, of which the effects are still to be observed in the future, shape international trade dynamics and influence corporate strategies. Hence, the global EV industry is at a critical

turning point, shaped by resource access and critical minerals' trade policies, battery technology advancements, the expansion of EV infrastructure, and government interventions. Looking into the long-run, the competition among these regions will greatly depend on technological advancements, trade policies, and access to critical resources (Network for Greening the Financial System, 2024). Not only the economic patterns are changing, but also the geopolitical implications are significant, as countries compete for resource security and supply chain control (Mehrotra, 2019), as will be later analysed more in depth.

2.3. Infrastructure and Consumers

The following insights further illustrate the complex interplay between developments in EV-supporting infrastructure, future EV production and demand increase, challenges faced for supply chains and infrastructure, and geopolitical implications.

The TRENDS Research & Advisory report (2024) highlights that despite supply chain disruptions for EV batteries and high commodity and electricity prices (especially after the breakout of the Russia-Ukraine war), EV sales have boosted, with record figures recorded in 2022, and forecasts remain bullish. Even if growth rates are slightly decelerating (as noted in McKerracher's analysis from BloombergNEF of 2024), overall EV sales are projected to reach record levels in the coming years, driven by strong demand in China, the US, and an evolving European market (Trendsresearch.org., 2024), (McKerracher, 2024). In Europe and in the US, the Electromobility Market report underscores a dynamic landscape where consumer registrations, model availability, and manufacturer market shares are rapidly evolving. This signals both a shift in consumer preference, as well as a competitive realignment among manufacturers as they compete to capture both domestic and global markets (Statista, 2025b), reflecting not only an evolving consumer behaviour, but is also a signal to manufacturers to redistribute their portfolios to stay competitive in this rapidly transforming industry (Statista, 2025c).

As previously mentioned, the governments and authorities of all three regions are working towards facilitating the growth of their share of the EV market, though there are some exogenous factors that they cannot fully control for, such as the availability and costs of raw materials involved for battery production. In fact, the cost of lithium-ion batteries has significantly dropped from \$806/kWh in 2013 to \$115/kWh in 2024, improving EV affordability, and simultaneously, battery recycling and second-life applications are projected to become a \$64 billion market by 2030, improving resource sustainability (Statista, 2025d). Policies should aimed at both stimulating EV adoption and ensuring secure supply chains for critical minerals, allowing for the first hypothesis (H1) to be postulated: ***H1: Government policies, including subsidies and tax incentives, significantly influence EV market dynamics by affecting production costs, consumer adoption, and cross-border investment flows.***

In order to support the diffusion of EVs, several EU initiatives and national mandates aim for thousands of public chargers (IEA, 2021a). China adopted a unified strategy by constructing the largest public charging network in the world to support its enormous EV production. According to a Statista analysis of China's EV charging infrastructure in 2025, the country had more than 2.7 million publicly available chargers in 2023, which, combined with cheaper charger costs than the US, demonstrates the competitive advantage that comes from strategic infrastructure investments (Statista, 2025a). Instead, the US's landscape on one side carries cohesive green policies that favour EV supply and adoption depending on its presidential mandate, while on the other hand, it still needs to invest in EV infrastructure minding China's advantage on price. In this way, the US could bridge the gap between the current range anxiety and supporting mass adoption of its domestic EVs. This need for charging infrastructure leads to the second hypothesis (H2): ***H2: The availability of robust EV charging infrastructure is a key factor driving regional EV adoption and trade performance.***

The TRENDS Research & Advisory report's (2024) additional economic analysis supports the EV transition's strategic economic stakes. It notes that even amidst global supply chain challenges and fluctuating commodity prices, the underlying economic rationale for the shift to EVs remains robust, driving both innovation and competitive restructuring in the automotive sector (Trendsresearch.org., 2024).

2.4. Trade Changes - CMs Versus Crude Oil

As previously mentioned, EV production relies heavily on critical minerals, which introduces challenges related to supply chain stability, international trade tensions, and price volatility. In fact, as the world transitions from fossil fuels, a key trend emerges: a shift in trade balances from crude oil, historically dominant in the global economy, to CMs, which are crucial for the production of EV batteries. This shift is reshaping the geopolitical landscape, especially among the US, the EU, and China, and thus the third hypothesis (H3) is formed: ***H3: Shifts in trade flows from crude oil to critical minerals are reshaping global supply chains and trade balances, impacting the international competitiveness of EV industries in the EU, US, and China.***

Furthermore, the fact that China dominates both mining and downstream processing poses geopolitical risks, and its export restrictions and regulatory measures can be used as a lever in trade disputes. The Critical Mineral Agreement (CMA) is a tool to reinforce the supply chains for CMs between countries as a response to the growing demand for EVs and consequently for the CMs used, given that the ownership and availability of CMs is not evenly distributed worldwide. Through the CMA, the EU aims at supporting circular economy principles and reducing environmental impacts, advocating for an agreement encompassing over 50 minerals and materials crucial for the green transition, not just as raw materials, but also obtained through recycling (Szczepański, 2023). To recap, the five critical minerals analysed in this research are lithium (Li), cobalt (Co), manganese (Mn), nickel (Ni), and graphite (C). Data will be analysed in the following sections, while the **Figure A1** in the

Appendix depicts a summary of the main locations of country reserves and production of CMs (Ritchie et al., 2024). This stated, a sudden spike in demand for these scarce resources (particularly caused by the green transition) may result in a major supply-demand mismatch because of the lengthy development cycles, which typically take more than ten years from exploration to production. This means that shortages and price increases may appear before new capacity can be implemented (IEA, 2021b).

The worldwide drive for cleaner energy is already causing a transformation in the oil and gas industries. The crude oil industry is predicted to remain financially significant at least until the next decade (IEA, 2023a), and in many of the leading scenarios depicted by the International Energy Agency's Net Zero Emissions by 2050 pathways, oil demand is projected to drop dramatically after 2030, accompanied by falling revenues (Chronis et al., 2023). Important to note, the pace of this transition will vary regionally, and it is highly likely that certain sectors (such as petrochemicals or aviation) may retain oil use longer: according to analysis in the literature, the timeframe for the complete transition away from fossil fuels may take until around 2050.

Overall, this research benefits from a multi-layered analysis: by comparing the three regions' capabilities and willingness to smoothen the scenario for the increase demand, production, and government support for EVs and the necessary complementary assets, it provides a comprehensive picture of the challenges and opportunities in the evolving EV market.

6. Conclusions and Recommendations

6.1. Final Conclusions

This study explores how the international trade for the market of electric vehicles between the EU, the US, and China varies, considering the change in trade flows of crude oil compared to CMs, the infrastructure for EVs, and the impact of government intervention on the EV market through regulations, monetary and fiscal policies, revealing significant insights into the growth of the EV market. Despite global supply chain challenges and fluctuating commodity prices, the underlying economic rationale for the shift to EVs remains robust, driving innovation and competitive restructuring in the automotive sector. Overall, this study can conclude that the findings highlight a multi-layered transformation of the automotive industry, where government intervention, resource access, and charging infrastructure are all necessary, yet individually insufficient conditions for competitiveness in the global EV trade landscape.

Policy-wise, it is challenging to establish ex-ante and ex-post effects of policies and regulations, as the effects of their implementation cannot be observed immediately, but rather their effects may take months or even years for them to have an actual impact. Also, policies that tackle EVs are highly linked to those that drive positive environmental impacts, thus these policies often change in function of the political administration in charge in a certain country at a given moment in time. The political reasons behind the choice of the policies are vastly driven by geopolitical reasons and market tensions, in order to drive domestic innovation and remain competitive worldwide. Therefore, this paper cannot conclude on which region among the EU, US, and China is using policies in the most effective way across all factors (consumer adoption, regulatory requirements, manufacturing costs, etc.) to increase their EV market growth, though China may be in a position of advantage given its substantial investments in fast charging infrastructure.

This mentioned, from policy analysis, an anticipation of when ICEVs will be substituted by EVs is risky. While regions can take measures such as improving their charging infrastructure and implementing taxes and subsidies to boost the domestic uptake and production of EVs, the trade and supply chains of critical minerals remain largely beyond their control. Therefore, this paper concludes that access to and stability of these supply chains will be a key factor to smoothen the transition from ICEVs to EVs.

6.2. Recommendations for the US, EU, and China

The US, EU, and China are each positioning themselves in the global critical minerals race to gain or maintain their market position in the EV industry. Currently, China holds a dominant position in many aspects of the supply chain, while the US and EU are seeking to catch up, thus each region should adopt strategies tailored to its strengths and weaknesses to remain competitive in the critical minerals era. Since one of the main strengths of the US lie on its technological developments and investments in R&D, it should build domestic capacity given its relative scarcity on raw materials and cultivate secure alliances for further support. The EU should follow up with its Memorandum Of Understanding (MOU) agreements through partnerships with resource-rich countries to foster their development by facilitating FDIs in these countries. Also, the EU could emerge as a leader in battery recycling requirements and systems, as its strong points are regulations for sustainable practices and environmental preservation. In turn, China sets the pace by a wide margin in supply chain control for CMs and cost competitiveness for EVs, having turned critical minerals into manufactured products at scale. However, it should focus on consolidating its gains whilst evolving towards a more sustainable and secure supply player, as the country's dominance has sometimes come with environmental costs (pollution from refining) and labour issues.

Altogether, whoever secures reliable access to lithium, cobalt, nickel, manganese, and graphite, and can produce them at scale responsibly, will hold the keys to the future energy

economy. The competitive advantage might thus be with China in the short- to medium-term for cost and capacity, whereas the US and EU aim to compete on innovation (better/substitute technologies or alternative materials) and ethical, sustainable production as selling points. Others regions like Australia, Canada, and some African nations (with EU/US support) will also play crucial roles, but the trio of US, EU, China largely shapes the strategic narrative: China as incumbent leader with pricing and scale advantages, and the US and EU striving to remain competitive also by investing in innovation and the development of their human capital. If these recommendations are pursued, the critical minerals supply landscape by 2030 will likely be more stable and diversified, reducing the risk of severe shortages and ensuring that the clean energy transition is not derailed by resource constraints. By 2050, all three regions (and the world) would benefit from a scenario where critical mineral supply is ample, responsibly produced, and equitably accessible, underpinning a successful move away from crude oil to a cleaner energy system.

6.3. Future Areas of Research

As a growing industry, future research can be conducted on the effects of the fast-paced developments of EV batteries, ranging from their reusability, vehicle-to-grid technologies, or more sustainable chemistries that also support a higher energy density and faster charging. Systems to transport electricity effectively and efficiently are still in a phase of development, thus investigations can also be conducted on how to improve such systems to guarantee a worldwide coverage to expand EV charging stations. Finally, since one of the main rationales behind electric vehicles is to decrease pollution from transportation, future research can also be directed to assessing the effective carbon-neutrality goals reached over time and the level to which electricity for EVs is sustainably produced.